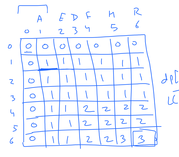
```cpp
int lcsUtil(int i, int j, string s1, string s2, vector<vector<int>> &memo) {
    if (i == s1.length() || j == s2.length()) {
        return 0;
    }
    if (memo[i][j] != -1) {
        return memo[i][j];
    }

    if (s1[i] == s2[j]) {
        memo[i][j] = 1 + lcsUtil(i + 1, j + 1, s1, s2, memo);
        return memo[i][j];
    }

    int memo1[i][j] = max(lcsUtil(i + 1, j, s1, s2, memo), lcsUtil(i, j + 1, s1, s2, memo));
    return memo1[i][j];
}
```



$dp[i][j] = LCS(S1[0 \dots i-1], S2[0 \dots j-1])$

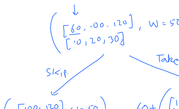
$LCS(A, AE) = \max(LCS(\text{"}\text{"}, AE), LCS(A, A))$

0-1 Knapsack

vals: [60, 100, 120]
wts: [10, 20, 30]

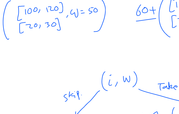
W = 50

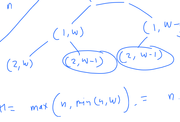
o/p: 220

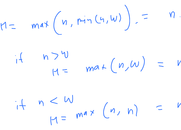


W = 50

o/p: 160







$M = \max(n, \min(n, W)) = n$

if $n > W$
$M = \max(n, W) = n$

if $n < W$
$M = \max(n, n) = n$

TC: $O(2^n)$
AS: $O(n)$